

CS-1872

Examination –May- 2024

B.Tech. VII Sem: Computer Science & Engineering
Cloud Computing

Time : 3 Hrs

Max. Marks : 70

Min. Marks : 22

: 70
: 22
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Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a – Max 50 words, Part b – Max 50 words,

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) Define distributed system.

Marks	BL	CO
02	1	1
02	2	1
03	1	1
07	1	1

(b) Define cloud computing and write main features of cloud computing.

(c) Explain limitations of cloud computing.

(d) Write short notes on the deployment and delivery of cloud service models with a neat diagram.

OR

Briefly summarize the Cloud Computing Reference Model. What is the major advantage of cloud computing? Explain.

07	2	1
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Q.2 (a) Define Virtualization. Explain need of virtualization in cloud computing?

02	2	2
02	2	2
03	1	2
07	2	2

(b) Define load balancing in cloud computing.

(c) List types of virtualization? Define each type of virtualization.

(d) What are the main characteristics of a Platform-as-a-Service solution? Describe the different categories of options available in a PaaS market.

OR

Explain types of Hypervisors. Explain architecture of Hyper-V. Discuss its use in cloud computing

07	2	2
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Q.3 (a) Write brief note on dark web

02	1	3
02	1	3
03	1	3
07	3	3

(b) Define AWS. What types of services does it provide?

(c) Write brief note on Amazon Web Service components.

(d) What type of service is AppEngine. Describe the core components of AppEngine. List major features of Google App Engine service.

OR

Describe Amazon EC2. Explain EC2 Components and its features.

07	1	3
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- Q.4 (a) What is AppFabric and which services does it provide?
 (b) Write brief note on Windows Live services.
 (c) What is SQL Azure? Illustrate the architecture of SQL Azure.
 (d) What is Windows Azure? Describe the architecture of Windows Azure.

02	1	4
02	1	4
03	2	4
07	2	4

OR

What is the Windows Azure Platform Appliance? For which kinds of scenarios was this appliance designed?

07	1	4
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- Q.5 (a) Define Brokered cloud storage access.
 (b) Write in brief service attributes in cloud.
 (c) What are the key mechanisms for protecting data.
 (d) Explain Protocol stack for an SOA architecture

02	1	5
02	1	5
03	2	5
07	1	5

OR

When you are deploying a cloud the risks in any cloud deployment are dependent upon the particular cloud service model chosen and the type of cloud on which you deploy your applications. In order to evaluate your risks, what analysis you need to perform? Explain.

07	1	5
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Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

AI-2053
Examination –Dec- 2023
B.Tech. V Sem : AIADS
Cloud Computing

Max. Marks : 70
 Min. Marks : 22

Time : 3 Hrs

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			Marks	BL	CO
Q.1	(a)	What is cloud computing? Write its key characteristics.	02	1	1
	(b)	Discuss deployment and service model.	02	2	1
	(c)	Differentiate between Hypervisor Type 1 and Type2.	03	2	1
	(d)	Explain the NIST architecture of cloud with diagram.	07	2	1
OR					
		Discuss the four dimensions of cloud cube model.	07	2	1
Q.2	(a)	What are the two formats of virtual appliances?	02	1	2
	(b)	What additional features does ADC provide over Load balancer?	02	2	2
	(c)	How does virtualization work?	03	3	2
	(d)	Discuss the various network resources in cloud computing in brief.	07	2	2
OR					
		Discuss migration and its types.	07	2	2
Q.3	(a)	What is GCP?	02	1	3
	(b)	What is Google Indexing and how does it work?	02	3	3
	(c)	Differentiate between Surface Web, Deep Web and Dark web.	03	2	3
	(d)	Discuss Amazon EC2, its features and use cases.	07	2	3
OR					
		Discuss Amazon S3, its features and use cases.	07	2	3

Q.4	(a)	What are region pairs in Microsoft Azure?	02	1	4
	(b)	Differentiate between CapEx and OpEx with an example.	02	2	4
	(c)	Explain the physical infrastructure of Microsoft Azure.	03	2	4
	(d)	What are the features of SQL server? Draw a diagram to represent the separation of responsibilities for Azure SQL server.	07	1	4

OR

		What are the features of Azure SQL data base? Draw a diagram to represent the separation of responsibilities for Azure SQL data base.	07	1	4
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Q.5	(a)	What are the types of SLA?	02	1	5
	(b)	What are the legal issues in cloud computing?	02	2	5
	(c)	Write short note on common change management models.	03	2	5
	(d)	Explain About the Cloud, Digital Identity, And Data Security.	07	1	5

OR

		Explain About Data Privacy and Security Issues.	07	1	5
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			Marks	BL	CO
Q.1	(a)	Define distributed system. Write its characteristics.	02	1	1
	(b)	Define cloud computing and Identify the main features of cloud computing.	02	2	1
	(c)	Explain limitations of cloud model.	03	1	1
	(d)	Explain the different types of models in the cloud computing .Explain the three cloud services in cloud computing.	07	1	1
OR					
		What does Infrastructure-as-a-Service refer to? What are the basic components of an IaaS-based solution for cloud computing? Provide some examples of IaaS implementations.	07	2	1
Q.2	(a)	Define Virtualization. Write advantages of virtualization?	02	2	2
	(b)	Write importance of load balancing in cloud computing.	02	2	2
	(c)	Write benefits of virtualization in the context of cloud computing.	03	1	2
	(d)	What does the acronym SaaS mean? Write Distinction between SaaS and PaaS. Give the name of some popular Software-as-a-Service solutions.	07	2	2
OR					
		Describe different types of Hypervisors with block diagram. Discuss its use in cloud computing.	07	2	2
Q.3	(a)	Give some name of Google API.	02	1	3
	(b)	What types of services does AWS provide?	02	1	3
	(c)	Define Google Indexed search method.	03	1	3

- (d) What is AppEngine. Discuss the computer services offered by AppEngine. 07
List major features of Google App Engine service.

Marks : 70
Marks : 22
on has 4
Assume

OR

Describe Amazon EC2. Explain EC2 Components and its basic features. 07

- Q.4 (a) Differentiate between Microsoft Azure and AWS. 02 2
(b) List the various web applications can be deployed with Azure. 02 1 4
(c) What is SQL Azure? Illustrate the architecture of SQL Azure. 03 2 4
(d) Explain in brief about Windows Azure Platform Appliance. For which kinds of scenarios was this appliance designed? 07 2 4

OR

What is Windows Azure? Describe the architecture of Windows Azure. 07 1 4

- Q.5 (a) Define cloud service boundary. 02 1 5
(b) What are the key mechanisms for protecting data? 02 1 5
(c) What are factors involved in deciding to move an application to the cloud. 03 2 5
(d) Define service oriented architecture and also explain Protocol stack for an SOA architecture. 07 1 5

OR

Write short note on :

- (a) Cloud mail service
(b) Cloud bursting

07 1 5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

IoT-2064
Examination –April/May- 2023
B.Tech. VI Sem : Internet of Things
Introduction to Cloud Computing for IoT

Time : 3 Hrs

Max. Marks : 70

Min. Marks : 22

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- Q.1 (a) Define grid computing. 02
 (b) How Cloud Computing Differ from Cluster Computing? 02
 (c) What is storage networking? 03
 (d) Discuss the operational and economic benefits of SaaS. 07

OR

How does enhance cloud computing environments using a cluster as a service? 07

- Q.2 (a) What is autonomic computing? 02
 (b) What is thin client? 02
 (c) State the limitations of virtualization. 03
 (d) Cloud services span the gamut of computer applications: audio and video streaming, instant messaging, and so forth. How they can work on cloud platform? 07

OR

Scalability is a characteristic of cloud computing which is used to handle the increasing workload by increasing in proportion amount of resource capacity. What is the difference between scalability and elasticity? 07

- Q.3 (a) Define hybrid cloud. 02
 (b) What is SPI? 02
 (c) Explain Google App Engine service. 03
 (d) Explain About Virtual Machines Provisioning and Manageability. 07

OR

Cloud services or run entirely in the cloud then, Explain the evolution of cloud computing and its challenges. 07

- Q.4 (a) What is Content Delivery Network? 02
 (b) Explain Cloud Management. 02

- (c) Explain About Cluster As A Service: The Logical Design.
- (d) Explain in detail the various aspects for the need of virtualization in cloud computing.

OR

How some threats to virtualized systems do are general in nature, as they are inherent threats to all computerized systems? 07

Q.5 (a) Define life cycle of SLA. 02

(b) Explain data security in the cloud. 02

(c) Explain About The Cloud, Digital Identity, And Data Security. 03

(d) Explain virtualization security management. 07

OR

Online content that isn't indexed by search engines belongs to what has come to be called the "Deep Web"—that is, content on the World Wide Web that is hidden. What is this? Explain in Detail. 07

MCA-304
Examination – Dec- 2023
MCA III Sem
Cloud Computing

Time : 2 Hrs

Max. Marks : 60

Min. Marks : 24

Note: Total number of questions are 10. Attempt any one question (Including all part) from each unit. Assume missing data, if any, suitably.

UNIT-I			
Q.1	(a)	Define cloud computing and Identify the main features of cloud computing.	06
	(b)	Explain the different types of models in the cloud computing with example.	06
OR			
Q.2	(a)	What is Windows Azure? Describe the architecture of Windows Azure.	06
	(b)	What does Infrastructure-as-a-Service refer to? What are the basic components of an IaaS-based solution for cloud computing? Provide some examples of IaaS implementations.	06
UNIT-II			
Q.3	(a)	What is SOP? Explain importance of service oriented architecture.	06
	(b)	Define Virtualization. What are the advantages of virtualization? List and discuss various virtualization types.	06

OR

		Identify the purpose of Differential and linear cryptanalysis and explain with neat diagram.	07		
			02	01	
Q.4	(a)	Define public key and private key.	02	02	
	(b)	Illustrate why is asymmetric cryptography bad for huge data.	03	04	4
	(c)	Explain attacks of RSA cryptosystem.			
	(d)	Perform encryption and decryption using RSA algorithm for $p=17$, $q=11$, $e=7$, $m=88$.	07	4	4

OR

		Explain Diffie-Hellman key exchange algorithm with its merits and demerits.	07	4	4
			02	3	5
Q.5	(a)	Explain three requirements for authentication.	02	1	3
	(b)	Define properties a digital signature.			
	(c)	Illustrate the security services provided by digital signature.	03	2	5
	(d)	Illustrate how Hash function algorithm is designed? Explain their features and properties.	07	2	5

OR

		Illustrate the steps involved in Signature generation and Verification functions of DSS.	07	2	5
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Q.1	(a)	Define distributed system. Write its characteristics.	02	1	1
	(b)	Define cloud computing and Identify the main features of cloud computing.	02	2	1
	(c)	Explain limitations of cloud model.	03	1	1
	(d)	Explain the different types of models in the cloud computing .Explain the three cloud services in cloud computing.	07	1	1
OR					
		What does Infrastructure-as-a-Service refer to? What are the basic components of an IaaS-based solution for cloud computing? Provide some examples of IaaS implementations.	07	2	1
OR					
Q.2	(a)	Define Virtualization. Write advantages of virtualization?	02	2	2
	(b)	Write importance of load balancing in cloud computing.	02	2	2
	(c)	Write benefits of virtualization in the context of cloud computing.	03	1	2
	(d)	What does the acronym SaaS mean? Write Distinction between SaaS and PaaS. Give the name of some popular Software-as-a-Service solutions.	07	2	2
OR					
		Describe different types of Hypervisors with block diagram. Discuss its use in cloud computing.	07	2	2
OR					
Q.3	(a)	Give some name of Google API.	02	1	3
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Describe Amazon EC2. Explain EC2 Components and its basic features. 07

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|-----|-----|--|----|---|---|
| Q.4 | (a) | Differentiate between Microsoft Azure and AWS. | 02 | 2 | |
| | (b) | List the various web applications can be deployed with Azure. | 02 | 1 | 4 |
| | (c) | What is SQL Azure? Illustrate the architecture of SQL Azure. | 03 | 2 | 4 |
| | (d) | Explain in brief about Windows Azure Platform Appliance. For which kinds of scenarios was this appliance designed? | 07 | 2 | 4 |

OR

What is Windows Azure? Describe the architecture of Windows Azure. 07 1 4

- | | | | | | |
|-----|-----|---|----|---|---|
| Q.5 | (a) | Define cloud service boundary. | 02 | 1 | 5 |
| | (b) | What are the key mechanisms for protecting data? | 02 | 1 | 5 |
| | (c) | What are factors involved in deciding to move an application to the cloud. | 03 | 2 | 5 |
| | (d) | Define service oriented architecture and also explain Protocol stack for an SOA architecture. | 07 | 1 | 5 |

OR

Write short note on : 07 1 5
(a) Cloud mail service
(b) Cloud bursting

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